

Reconciling the Delphes $\tau_h\tau_h$ channel with CMS NanoAOD

A cutflow diagnostic, six corrections, and what is still open

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Abstract

The tuned Delphes $HH \rightarrow b\bar{b}\tau^+\tau^-$ sample initially selected $\sim 5\times$ *fewer* $\tau_h\tau_h$ events than the CMS NanoAOD reference, and its $m_{\tau\tau}$ peaked ~ 25 GeV high. A five-stage cutflow localised the yield deficit to a single stage — the FastMTT di- τ mass — and the cause turned out not to be a detector or tuning effect but a representational one: **in Delphes a τ_h is an AK4 jet, whereas in CMS it is an HPS τ_h object**. Nearly every discrepancy we found traces to that one difference, expressed in a different component of the four-vector or in a different reference used to tune it. Six corrections are described, each anchored to the CMS NanoAOD and applied downstream so that no Delphes re-production is required. The $\tau_h\tau_h$ yield ratio moved from 0.20 to 1.25 and the $m_{\tau\tau}$ peak offset from ~ 25 to ~ 7 GeV; seven of the ten NSBI input features now agree with CMS within statistics. Two discrepancies remain open. This note is circulated for cross-check.

1 Motivation and summary

The Delphes fast-simulation sample must reproduce the CMS $HH \rightarrow b\bar{b}\tau\tau$ Run-3 baseline closely enough to support a phenomenological unbinned NSBI κ_λ measurement. The $b\bar{b}$ side and the κ_λ -carrying observable m_{HH} agreed from early on; the $\tau\tau$ side did not, in two ways that resisted object-level tuning: a factor ~ 5 yield deficit and a ~ 25 GeV shift in $m_{\tau\tau}$.

Table 1 is the summary. Each row is one correction, all of them downstream of the Delphes card (the production is unchanged), and each was adopted only after a measurement identified it.

#	correction	Delphes N	Delphes/CMS	$m_{\tau\tau}$ peak offset
0	(starting point, symmetric selection)	289	0.20	+25
1	τ_h visible <i>mass</i> from the anchor	2590	1.81	+25
2	hadronic vs leptonic gen τ	2100	1.47	+25
3	common τ -candidate acceptance	1729	1.30	+25
4	τ energy scale anchored to CMS	1721	1.30	+25
5	p_T^{miss} smearing (pileup)	1721	1.30	+12
6	visible- τ energy reference	1660	1.25	$\sim +7$

Table 1: Cumulative effect of the six corrections, at $\kappa_\lambda = 0$ from 2×10^4 generated events (CMS reference: 1431 events, falling to 1328 once the common τ acceptance is applied to both sides). Peak offsets are read off the normalised histograms and are approximate.

2 The diagnostic

Both samples are walked through the same five stages; a stage whose *conditional* efficiency differs between the two columns is where a difference is created.

Two readings matter. First, **stage 5 alone** carried the deficit: Delphes retained 12% where CMS retained 99%, and through stage 4 Delphes was in fact *ahead* in cumulative efficiency. Second, the large stage-1/2 ratios were an artefact of the diagnostic, not of the simulation: the Delphes visible- τ proxy could not distinguish a hadronic from a leptonic τ decay, so the denominator was inflated by roughly $1/\mathcal{B}(\tau_h\tau_h)$. Correction 2 removed it.

#	stage	Delphes	CMS NanoAOD
1	gen	≥ 2 visible hadronic gen τ	≥ 2 GenVisTau
2	reco	≥ 2 accepted jets matched to those	≥ 2 Tau objects matched
3	ID	≥ 2 passing the re-tagged TauTag	≥ 2 passing DeepTau v2.5 VSjet Medium
4	jets	≥ 2 jets surviving the overlap removal	identical
5	$m_{\tau\tau}$	FastMTT mass finite and > 20 GeV	identical

Table 2: Cutflow stages. τ candidates $p_T > 20$ GeV, $|\eta| < 2.3$; jets $p_T > 20$ GeV, $|\eta| < 2.4$; matching and overlap removal $\Delta R < 0.4$.

stage	initial			after correction 1			after correction 2		
	$\varepsilon_{\text{step}}$	D/CMS	$\Delta\varepsilon$	$\varepsilon_{\text{step}}$	D/CMS	$\Delta\varepsilon$	$\varepsilon_{\text{step}}$	D/CMS	$\Delta\varepsilon$
1 gen visible τ_h	0.499	/ 0.235	2.12	0.499	/ 0.235	2.12	0.274	/ 0.235	1.17
2 reco τ_h object	0.635	/ 0.855	0.74	0.635	/ 0.855	0.74	0.946	/ 0.855	1.11
3 τ_h ID	0.370	/ 0.386	0.96	0.370	/ 0.386	0.96	0.373	/ 0.386	0.97
4 cleaned jets	0.865	/ 0.885	0.98	0.865	/ 0.885	0.98	0.870	/ 0.885	0.98
5 FastMTT $m_{\tau\tau}$	0.120	/ 0.991	0.12	1.000	/ 0.991	1.01	0.999	/ 0.991	1.01
CMS/Delphes yield	5.60			0.67			0.81		

Table 3: Cutflow at $\kappa_\lambda = 1$, 2×10^4 events. $\Delta\varepsilon$ is the Delphes/CMS ratio of the per-stage efficiency. Stage 5 alone carried the entire initial deficit; stages 1–2 were an artefact of the diagnostic itself, removed by correction 2.

3 One root cause, several components

A CMS τ_h is an HPS object built from a narrow set of decay products. A Delphes τ_h is an anti- k_T $R = 0.4$ jet that happens to carry **TauTag**. The two therefore differ in *every* component of the four-vector, and additionally in the generator-level reference one would naturally use to tune each component:

- **mass** — the jet mass is $\mathcal{O}(5\text{--}20)$ GeV; a τ_h visible mass is bounded by $m_\tau = 1.777$ GeV (correction 1);
- **energy** — the jet carries the underlying event inside the cone, so it is $\sim 17\%$ harder than the visible τ at low p_T (corrections 4 and 6);
- **reference** — a Delphes **GenJet** is also a jet and also contains the underlying event, so it is *not* the analogue of **GenVisTau** (correction 6);
- **identity** — “matched to a gen τ ” does not mean “hadronic” (correction 2), and the two data tiers expose τ candidates with different intrinsic thresholds (correction 3).

Only correction 5 (p_T^{miss}) is unrelated to this; it follows from the card running without pileup by design.

3.1 Correction 1: the τ_h visible mass, and why it must be sampled

The covariance-free FastMTT estimator maximises, per event, a likelihood over the visible energy fractions $x_1, x_2 \in (0, 1]$, with $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ and a hadronic decay prior

$$\mathcal{L}_{\text{dec}}(x) = \begin{cases} 1, & x_{\min} \leq x \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad x_{\min} \equiv (m_{\text{vis}}/m_\tau)^2. \quad (1)$$

If $m_{\text{vis}} > m_\tau$ then $x_{\text{min}} > 1$, the admissible interval is empty, the likelihood vanishes identically and the estimator returns NaN. A Delphes τ -jet carrying its jet mass therefore admits *no solution at all*: $\sim 89\%$ of $\tau_h\tau_h$ pairs were silently lost, which was the whole of the stage-5 collapse.

The correction measures the CMS τ_h visible mass on the anchor and assigns it to τ -tagged Delphes jets. Crucially it must be **sampled, not averaged**. An initial implementation assigned the per- p_T median; that is wrong because via Eq. (1) the mass is a one-sided *lower bound* on x , not a smearing. Its contribution to the numerator m_{vis} is negligible ($m^2 \sim 1.6$ against $m_{\text{vis}}^2 \sim 3600 \text{ GeV}^2$), so a single value *relaxes* a real constraint for every leg whose true mass lies above it. For a 3-prong a_1 leg ($m_{\text{vis}} = 1.26$, floor $x_{\text{min}} = 0.503$) given the median 0.77 ($x_{\text{min}} = 0.188$), with p_T^{miss} favouring $x \simeq 0.25$, the fit moves from $x = 0.525$ to $x = 0.250$ and $m_{\tau\tau}$ from 224 to 470 GeV. Over a realistic decay-mode mixture the median assignment inflates the $m_{\tau\tau}$ width by $\sim 60\%$ and the mean by $\sim 9\%$ while leaving the *mode* unchanged — so the $Z \rightarrow \tau\tau$ candle, which gates the mode, would not have caught it. The implementation therefore stores the anchor’s per- p_T mass *quantiles* and draws per τ -jet from them.

3.2 Corrections 2–3: identity and acceptance

The τ_h efficiency map is measured on the anchor as **GenVisTau** $\rightarrow$ DeepTau-Medium Tau, and **GenVisTau** exists only for hadronic decays. Treating every gen τ as genuine handed a leptonic τ ’s jet that efficiency (~ 0.5) instead of the jet $\rightarrow \tau_h$ fake rate (~ 0.004), fabricating $\tau_h\tau_h$ events out of $\tau_h\tau_\ell$ and $\tau_\ell\tau_\ell$ ones. A gen τ is now classified by looking for a status-1 e/μ whose first non-self-copy ancestor is a τ , using the generator mother links only.

Separately, the two tiers expose τ candidates with different intrinsic floors: a Delphes τ_h is a jet and so exists down to the card’s **JetPMin** of 15 GeV and out to $|\eta| < 2.5$, while the CMS Tau collection begins near 18–20 GeV. A common acceptance ($p_T > 20$, $|\eta| < 2.3$, the analysis values) is now applied to both sides before the pair is chosen.

3.3 Corrections 4 and 6: the τ energy scale and its reference

The energy-scale map was originally derived by inverting the Delphes response to unity against its own **GenJet** — a *self*-comparison, where every other map in the framework is a comparison *to CMS*. Correction 4 replaced it with the anchor ratio, **tau_escale** = $\mathcal{R}_{\text{CMS}}/\mathcal{R}_{\text{Delphes}}$, in the same form already used for the lepton scale factors.

That alone changed almost nothing (1–3%), and the reason is instructive: the two responses were still measured against *different* references. The CMS response is **Tau/GenVisTau** — clean — while the Delphes response was τ -jet/**GenJet**, and a **GenJet** is an $R = 0.4$ cluster that still contains the underlying event (only *pileup* is absent from the card). Correction 6 builds a genuine visible gen τ on the Delphes side using the identity that a hadronic τ decay has exactly one neutrino,

$$p_{\text{vis}} = p_\tau - p_{\nu_\tau}, \quad (2)$$

with the ν_τ located through the same mother-link walk (no generator daughter links are needed). Profiled against *that*, the Delphes τ -jet is **17% harder than the visible τ** at 25 GeV, falling to $\sim 2\%$ at 250 GeV — the shape expected of a roughly fixed underlying-event offset. The resulting map is **tau_escale** = [0.85, 0.88, 0.90, 0.91, 0.93, 0.94, 0.94, 0.95], against [0.91, ..., 0.96] before.

3.4 Correction 5: p_T^{miss} resolution (pileup)

The card runs without pileup by construction, and its header states that “pileup enters through the tuning maps”. That map had never been built. Measured with the same estimator on both sides, the per-component p_T^{miss} resolution is 16.5 GeV on Delphes against 32.6 GeV on the CMS anchor — a factor of two, and the largest single mismatch anywhere in the map set. Gaussian noise of the quadrature difference (28.1 GeV) is added per component. Since p_T^{miss} is what FastMTT

fits the energy fractions against, this is the correction that moved $m_{\tau\tau}$ most: the peak offset fell from ~ 25 to ~ 12 GeV and the distribution broadened from far-too-narrow into the right range.

4 What is still open

1. **$\Delta R_{\tau\tau}$ shoulder.** Delphes carries an excess at $\Delta R_{\tau\tau} \simeq 2.2\text{--}3.2$ that CMS does not. It has survived every correction above, which is informative: none of them touched the τ *direction* or the selected *population*. Large $\Delta R_{\tau\tau}$ corresponds to a low- p_T Higgs, hence soft τ .
2. **Residual yield excess** of $1.25\times$ and a $m_{\tau\tau}$ distribution still somewhat broader than CMS.

A single candidate explains both, and is our current working hypothesis. The downstream corrections are applied in the order (b-tag, τ_h tag, energy scale, mass), so the τ_h efficiency is drawn at the *raw* jet p_T — which correction 6 shows to be $\sim 17\%$ too high at low p_T . The efficiency rises steeply there ($0.39 \rightarrow 0.52$ between 25 and 35 GeV), so evaluating it $\sim 17\%$ too far to the right produces a systematic over-efficiency for soft τ : too many reconstructed pairs, concentrated exactly in the low- p_T , large- $\Delta R_{\tau\tau}$ population. Applying the energy scale *first* would make every subsequent map evaluate at a CMS-like p_T . This is falsifiable: if the hypothesis is right the excess should vanish preferentially at low τ p_T and take the shoulder with it; if the shoulder survives a corrected low- p_T efficiency it is genuinely angular, and the τ direction (jet axis versus HPS axis) is the next candidate.

Hypotheses tested and rejected. For the record, three explanations for the $m_{\tau\tau}$ shift were proposed and excluded by direct measurement: (i) *fake* τ_h — the gen-matched/fake split gives fake fractions of 5.7% (Delphes) against 5.0% (CMS), essentially equal, with both shape discrepancies living entirely in the gen-matched component; (ii) τ -pair selection — making the selection fully symmetric moved $\Delta R_{\tau\tau}$ but left $m_{\tau\tau}$ untouched; (iii) *a large CMS τ energy miscalibration* — the anchor response is 0.97–0.99, so CMS τ_h are calibrated to the visible τ to within a few percent.

Not addressed here. Pileup also affects the jet energy *resolution* (only the scale is corrected) and produces pileup jets that Delphes cannot make at all. Both act on the $b\bar{b}$ side, which currently agrees well with CMS, and the second cannot be fixed downstream in any case.

5 Points for cross-check

1. **Is the framing accepted?** The claim is that essentially the whole τ -side discrepancy follows from representing a τ_h as an AK4 jet, and that the remedy is to correct each component of that four-vector against a CMS-anchored, like-for-like reference.
2. **Is quantile sampling of the visible mass sufficient?** The spread is reproduced, but the drawn mass is independent of the τ energy and of the partner leg. Would sampling a decay mode per τ be preferable, so that mass, prong multiplicity and energy sharing stay correlated?
3. **Is isotropic p_T^{miss} smearing adequate?** Real p_T^{miss} resolution is anisotropic and scales with ΣE_T ; ours is round and flat. Delphes $m_{\tau\tau}$ is now marginally *broad*er than CMS, which is what one would expect from over-simplified noise.
4. **The order-of-application hypothesis** in Section 5 — does it look right, and is applying the energy scale first the correct remedy?
5. **Residual normalisation.** Is a $1.25\times$ yield excess acceptable for an unbinned NSBI study, given that the training uses shapes and the normalisation is fitted, or should it be closed before production?

Reproducing

All corrections live in `delphes-pipeline` and are applied downstream of the Delphes card; the existing production is unchanged. Re-derive the maps and remake the comparison with

```
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \  
    --output cards/tuning/maps_v1.json  
python scripts/tauh_cutflow.py --config config.v1.yml [...] # the diagnostic  
python scripts/nsbi_overlay.py --config config.v1.yml [...] \  
    --tautau-only --diagnostics
```

The cutflow's unit tests inject a loss at one known stage at a time and assert it is attributed to that stage and no other; the overlay's tests assert that the same physical object is accepted or rejected identically on both sides.